

Data Card — Bridging the Tokenization Gap in African LLMs via SuperBPE

Input vs. Output

Inputs: Raw text corpora in African languages (Amharic, Afaan Oromo, Tigrinya, Swahili, Hausa, Yoruba, Somali, etc.), collected from open datasets, community contributions, and public domain texts.

Outputs: Tokenized sequences produced by the SuperBPE tokenizer, optimized for language model training.

Purpose: Reduce tokenization bias toward English and improve inclusivity for African languages.

Classification Problem

This project is not a classification task. Instead, the tokenizer generates vocabulary merges and token sequences. Evaluation focuses on lossless round-trip encoding/decoding and minimizing emitted tokens.

Labeling

Source of labels: Algorithmic merges derived from token frequency statistics.

Quality checks:

- Round-trip fidelity tests (encode → decode).
- Cross-language consistency checks.
- Manual validation by native speakers for low-resource languages.

Data Creation

Collection strategy:

- Public multilingual corpora (JW300, OSCAR, Wikipedia dumps).
- Community-curated African language texts (oral histories, radio transcripts, local blogs).
- Synthetic augmentation for under-resourced languages.

Process: Unicode normalization, cleaning, whitespace handling, and merge training up to 20k vocab size.

Web Collection Ethics

Consent: Only voluntary community contributions are included.

Copyright: Restrict sources to open-licensed or public domain datasets.

Privacy: Anonymize sensitive or personal data.

Tools: Human curation plus preprocessing scripts; avoid scraping proprietary sources without permission.

Bias and Representation

Risk: English and Amharic corpora dominate available datasets.

Mitigation:

- Balance representation across languages.
- Actively include underrepresented voices (e.g., Oromo, Tigrinya, Somali).
- Monitor token distribution to avoid skew.

Goal: Fair tokenization reflecting Africa's linguistic diversity.

Values and Community Alignment

Values from Challenge 1: Inclusivity, fairness, cultural preservation, transparency.

Alignment:

- Diverse language coverage ensures equitable access to LLMs.
- Documentation and reproducibility promote accountability.

Community benefit: stronger representation of African languages in AI, empowering local researchers and users.

Dataset Summary (150–250 words)

Our dataset consists of multilingual African text corpora curated from open sources and community contributions. Inputs are raw text in languages such as Amharic, Afaan Oromo, Tigrinya, Swahili, Hausa, and Yoruba. Outputs are tokenized sequences generated by the SuperBPE tokenizer, which extends standard BPE by allowing merges across whitespace boundaries. This reduces emitted tokens while maintaining exact reversibility, enabling more efficient language model training. Data sources include public corpora (JW300, OSCAR, Wikipedia), community-curated texts, and synthetic augmentation for low-resource languages. Preprocessing ensures Unicode normalization, cleaning, and ethical handling of text. The dataset supports our project by bridging tokenization gaps in African LLMs, ensuring inclusivity and fairness across diverse languages. By balancing representation and mitigating bias, the dataset empowers African communities to build language models that reflect their linguistic and cultural diversity.

Ethics and Representation

We address privacy by excluding sensitive or personal data and anonymizing community contributions. Consent is respected by only using voluntary and open-licensed texts. Copyright is observed by restricting sources to public domain or permissively licensed datasets. Potential bias arises from the dominance of English and Amharic corpora; we mitigate this by deliberately including underrepresented languages and balancing token distribution. Fairness is improved through diverse language coverage,

ensuring that African LLMs do not disproportionately favor high-resource languages.

Values and Community Impact

Our dataset reflects values of inclusivity, fairness, cultural preservation, and transparency. It supports our values-led problem statement by directly addressing the tokenization gap in African languages. By curating diverse corpora and designing a tokenizer that respects linguistic diversity, we empower communities to build language models that serve their needs. Accountability is maintained through open documentation, reproducible pipelines, and community engagement. The impact is a more equitable AI ecosystem where African languages are represented fairly, strengthening cultural identity and enabling broader participation in AI development.

References

[1] Pushkarna, M., Zaldivar, A. and Kjartansson, O. 2022. Data Cards: Purposeful and Transparent Dataset Documentation for Responsible AI. In FAccT '22: ACM Conference on Fairness Accountability and Transparency. Seoul, Republic of Korea.